DHARMA FARMS

Software Requirements Document

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A comprehensive software solution for daily milk subscription management, delivery tracking, and customer communication via WhatsApp.

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1. EXECUTIVE SUMMARY

Dharma Farms operates a daily milk delivery service serving approximately 100 customers across multiple regions with a team of 5 delivery personnel. The business currently runs on manual processes including phone calls, handwritten ledgers, and informal WhatsApp messages, which creates operational inefficiencies, payment tracking gaps, and poor customer visibility.

This document outlines the requirements for a lightweight, WhatsApp-first software system designed specifically for small-scale daily delivery operations. The solution eliminates complexity while providing real-time tracking, automated notifications, and clear financial oversight.

Key Objectives:

- Replace manual customer tracking with unique ID-based digital records
- Enable real-time delivery status updates via WhatsApp (Traffic Light System)
- Automate subscription pause/resume with automatic day extensions
- Provide admin dashboard for payment tracking and operational oversight
- Require zero app installation for delivery boys and customers
- Maintain simplicity - no enterprise complexity, no unnecessary features

2. BUSINESS OVERVIEW

Parameter	Details
Business Name	Dharma Farms
Service Model	B2B and B2C Daily Milk Subscription Delivery
Customer Base	~100 active subscribers
Delivery Team	5 delivery boys (region-based allocation)
Delivery Window	5:00 AM - 7:00 AM daily
Service Radius	Limited geographic radius (local delivery)
Payment Model	Monthly Prepaid Subscription
Current Tools	Manual ledgers, phone calls, informal WhatsApp
Target Platform	WhatsApp-first (primary interface)

Current Pain Points:

- No centralized customer database - records scattered across notebooks and phones
- Payment tracking is manual and error-prone; difficult to identify overdue accounts
- No visibility into delivery status until boys report back hours later
- Customer skip requests handled ad-hoc via phone calls, leading to missed deliveries or double deliveries
- No automated way to notify customers of delivery status, arrival, or issues
- Delivery route planning done mentally - no optimization or tracking
- Admin lacks real-time operational dashboard; decisions based on delayed information

3. STAKEHOLDERS & USER ROLES

3.1 Admin (Business Owner / Manager)

- Full system access and control
- Manages customer onboarding, subscription assignments, and payment records
- Views daily dispatch board and real-time delivery progress
- Receives automated alerts for payment issues, delivery failures, and anomalies
- Generates monthly revenue and operational reports
- Manages delivery boy assignments and route allocations

3.2 Delivery Boy

- Receives daily route list via WhatsApp each morning
- Updates delivery status per customer via simple WhatsApp commands
- Reports issues (customer not home, address problems, etc.) in real-time
- No app installation required - operates entirely through WhatsApp
- Receives next customer details automatically after each status update

3.3 Customer

- Receives automated delivery status notifications via WhatsApp
- Can pause/resume subscription via WhatsApp commands (before 5:00 AM cutoff)
- Can check subscription balance and remaining days
- Receives confirmation of pause with automatic day extension
- No app installation required - operates entirely through WhatsApp

4. FUNCTIONAL REQUIREMENTS

4.1 Customer Management

ID	Requirement	Priority	Description
CM-01	Unique Customer ID	High	Every customer assigned a unique code (e.g., C001, C002)
CM-02	Customer Profile	High	Name, phone number, physical address, region/zone
CM-03	Customer Onboarding	High	Admin adds customer; system sends welcome WhatsApp message v
CM-04	Customer Search	Medium	Admin can search by name, phone, or customer ID
CM-05	Customer Status	High	Active / Paused / Inactive status tracking
CM-06	Region Assignment	High	Each customer assigned to a delivery zone and specific delivery boy

4.2 Subscription & Billing

ID	Requirement	Priority	Description
SB-01	Monthly Prepaid Plans	High	Customers pay in advance for 30-day subscription cycles
SB-02	Day Extension on Skip	High	If customer skips a day, subscription extends by +1 day automaticall
SB-03	Skip Cutoff Time	High	Skip/pause requests must be made before 5:00 AM for same-day eff
SB-04	Balance Tracking	High	System tracks remaining days in active subscription
SB-05	Payment Recording	High	Admin records cash/UPI payments; system updates customer ledger
SB-06	Payment Due Alerts	High	Automated notification to admin when customer approaches renewa
SB-07	Subscription History	Medium	Full history of subscriptions, skips, and extensions per customer

4.3 Delivery Operations

ID	Requirement	Priority	Description
DO-01	Traffic Light Status	High	Green=Delivered, Yellow=Arriving/Issue, Red=Skipped/Cancelled
DO-02	Daily Route Assignment	High	System auto-assigns customers to delivery boys by region
DO-03	Morning Dispatch	High	Admin receives auto-generated dispatch summary at 6:00 AM
DO-04	Real-time Status Updates	High	Delivery boys update status per customer via WhatsApp commands
DO-05	Auto Next Customer	Medium	After each update, system replies with next customer details
DO-06	Issue Reporting	High	Boys can report problems (not home, wrong address, etc.) with reas
DO-07	End-of-Route Summary	Medium	Delivery boy sends 'FINISH' - admin gets completion report
DO-08	Skip Auto-Extension	High	Red status automatically triggers +1 day subscription extension

4.4 WhatsApp Communication

Delivery Boy Commands:

```
DONE [CODE] - Mark customer as delivered (Green)
SKIP [CODE] - Customer not available, auto +1 day (Red)
ISSUE [CODE] [reason] - Report problem (Yellow)
ARRIVING [CODE] - Notify customer of arrival (Yellow)
FINISH - End route, send admin summary
```

Customer Commands (Before 5:00 AM Cutoff):

```
PAUSE [N] - Pause for N days, auto +N day extension
PAUSE [date] to [date] - Pause date range
RESUME - Resume from tomorrow
SKIP TOMORROW - Skip tomorrow only, +1 day
BALANCE - Check remaining subscription days
STATUS - Check today's delivery status
```

Admin Notifications:

- Daily dispatch summary at 6:00 AM
- Real-time delivery completion per boy
- Payment due alerts (within 1 hour of detection)

- Issue alerts from delivery boys
- End-of-day operational summary

4.5 Payment Tracking

ID	Requirement	Priority	Description
PT-01	Payment Recording	High	Admin records each payment with date, amount, mode (Cash/UPI)
PT-02	Customer Ledger	High	Running balance of paid vs. delivered days per customer
PT-03	Due Date Tracking	High	System calculates and alerts when subscription renewal is due
PT-04	Payment History	Medium	Complete payment history per customer with dates and amounts
PT-05	Monthly Revenue Report	Medium	Auto-generated monthly collection summary
PT-06	Overdue Alerts	High	Flag customers with expired subscriptions who still receive milk

5. NON-FUNCTIONAL REQUIREMENTS

Category	Requirement	Details
Performance	Response Time	WhatsApp replies within 3 seconds
Performance	Concurrent Users	Support 5 delivery boys + 100 customers simultaneously
Reliability	Uptime	99.5% uptime during delivery hours (5 AM - 8 AM)
Reliability	Data Backup	Daily automated database backup
Usability	Learning Curve	Delivery boys trained in under 10 minutes
Usability	Interface	WhatsApp-only for field staff; web dashboard for admin
Security	Data Protection	Customer phone numbers and addresses encrypted at rest
Security	Access Control	Role-based access (Admin vs. Delivery vs. Customer)
Scalability	Growth Ready	Architecture supports up to 500 customers without rework
Maintainability	Code Quality	Clean, documented codebase for future enhancements

6. TECHNICAL ARCHITECTURE

6.1 Recommended Stack

Layer	Technology	Rationale
Backend	Node.js + Express	Fast, lightweight, excellent WhatsApp API support
Database	PostgreSQL	Reliable, free, handles relational data (customers, subscripti
WhatsApp	WhatsApp Business API (Meta)	Official, reliable, template messages, scalable
Admin Dashboard	React.js + Tailwind CSS	Modern, responsive, fast to build
Hosting	AWS / DigitalOcean	Cost-effective, reliable, easy to scale
Scheduler	node-cron	Daily dispatch, backup, report generation
Notifications	WhatsApp API Webhooks	Real-time inbound message processing

6.2 System Architecture Overview

The system follows a simple three-tier architecture optimized for small business operations:

- Presentation Layer: WhatsApp (primary), Web Admin Dashboard (secondary)
- Application Layer: Node.js API server handling business logic, message parsing, and scheduling
- Data Layer: PostgreSQL database storing customers, subscriptions, deliveries, payments, and logs
- Integration Layer: WhatsApp Business API via Meta Cloud for all messaging
- Automation Layer: Scheduled jobs for daily dispatch, payment reminders, and reporting

7. DATA MODEL (Simplified)

Core Entities:

CUSTOMER

```
customer_id (PK), name, phone, address, region_id, status, created_at
```

SUBSCRIPTION

```
sub_id (PK), customer_id (FK), start_date, end_date, total_days,
remaining_days, status
```

DELIVERY

```
delivery_id (PK), customer_id (FK), delivery_date, status,
delivery_boy_id, timestamp
```

PAYMENT

```
payment_id (PK), customer_id (FK), amount, payment_date, mode,
subscription_id (FK)
```

DELIVERY_BOY

```
boy_id (PK), name, phone, region_id, status
```

REGION

```
region_id (PK), name, description
```

SKIP_LOG

```
skip_id (PK), customer_id (FK), skip_date, reason, extension_applied
```

ADMIN_USER

```
admin_id (PK), name, phone, email, role, password_hash
```

8. KEY WORKFLOWS

8.1 Daily Delivery Workflow

1. 05:30 AM - System generates daily delivery list per region
2. 06:00 AM - Admin receives dispatch summary via WhatsApp
3. 06:00 AM - Each delivery boy receives their route list via WhatsApp
4. 06:15 AM - Delivery boys begin routes
5. Per Customer - Boy sends 'DONE C001' -> System logs Green -> Sends confirmation to boy + customer
6. Per Customer - Boy sends 'SKIP C001' -> System logs Red -> Extends subscription +1 day -> Notifies customer
7. Per Customer - Boy sends 'ISSUE C001 reason' -> System logs Yellow -> Alerts admin immediately
8. 07:30 AM - Boy sends 'FINISH' -> Admin receives route completion summary
9. 08:00 AM - System generates daily report for admin review

8.2 Customer Pause Workflow

1. Customer sends 'PAUSE 3' via WhatsApp (before 5:00 AM)
2. System validates cutoff time -> If valid, processes request
3. System updates subscription: remaining_days + 3, status = PAUSED
4. System sends confirmation: 'Paused for 3 days. Resume Jan 8. +3 days added.'
5. Delivery boy's route auto-excludes customer for paused days
6. If after 5:00 AM -> System replies: 'Changes apply from tomorrow. Today's delivery confirmed.'

9. IMPLEMENTATION ROADMAP

Phase	Duration	Deliverables	Priority
Phase 1: Foundation	Week 1-2	Database setup, WhatsApp API integration, basic message handling	Critical
Phase 2: Core Features	Week 3-4	Customer management, subscription engine, delivery status tracking	Critical
Phase 3: WhatsApp Flows	Week 5-6	Delivery boy commands, customer commands, admin notifications	Critical
Phase 4: Admin Dashboard	Week 7-8	Web dashboard for admin: customers, payments, reports, dispatch view	High
Phase 5: Testing & Training	Week 9	UAT with delivery boys, customer onboarding, bug fixes	High
Phase 6: Go-Live	Week 10	Full production deployment, monitoring, support	Critical
Phase 7: Enhancement	Ongoing	Analytics, route optimization, B2B features, customer feedback loop	Medium

10. BUDGET & COST ESTIMATE

10.1 Development Costs (One-Time)

Item	Description	Est. Cost (INR)
Backend Development	Node.js API, database, business logic	**Rs.80,000 - Rs.1,20,000**
WhatsApp Integration	Meta API setup, webhook handling, message flows	**Rs.40,000 - Rs.60,000**
Admin Dashboard	React web app for admin management	**Rs.50,000 - Rs.80,000**
Testing & QA	UAT, bug fixing, performance testing	**Rs.20,000 - Rs.30,000**
Deployment & Setup	Server setup, SSL, domain, initial configuration	**Rs.15,000 - Rs.25,000**
Documentation & Training	User guides, delivery boy training	**Rs.10,000 - Rs.15,000**
Contingency (15%)	Unexpected requirements or delays	**Rs.30,000 - Rs.50,000**
TOTAL DEVELOPMENT		**Rs.2,45,000 - Rs.3,80,000**

10.2 Recurring Costs (Monthly)

Item	Description	Est. Cost (INR/Month)
Server Hosting	AWS/DigitalOcean VPS (2GB RAM, 50GB SSD)	**Rs.1,500 - Rs.2,500**
WhatsApp API	Meta Business API (~3,000 conversations/month)	**Rs.1,500 - Rs.3,000**
Database Hosting	Managed PostgreSQL or self-hosted	**Rs.500 - Rs.1,000**
Domain & SSL	Annual cost amortized monthly	**Rs.100 - Rs.200**
Backup Storage	Automated daily backups	**Rs.200 - Rs.400**
Maintenance & Support	Bug fixes, minor updates (optional retainer)	**Rs.5,000 - Rs.10,000**
TOTAL MONTHLY		**Rs.8,800 - Rs.17,100**

11. RISK ASSESSMENT

Risk	Impact	Likelihood	Mitigation
WhatsApp API Ban	High	Low	Use official Meta API; avoid unofficial tools
Delivery Boy Adoption	Medium	Medium	Extremely simple commands; voice note fallback; training
Internet Connectivity	Medium	High	Messages queue offline; boys can call admin as backup
Data Loss	High	Low	Daily automated backups; cloud redundancy
Scope Creep	Medium	High	Strict MVP focus; defer non-essential features to Phase 7
Customer Resistance	Low	Medium	Gradual rollout; keep phone call option during transition
Payment Tracking Errors	High	Medium	Double-entry validation; admin confirmation for all payments
Server Downtime	High	Low	Reliable hosting; monitoring alerts; quick recovery plan

12. APPENDICES

Appendix A: Sample WhatsApp Messages

Admin Morning Dispatch:

```
Good morning! Today's dispatch: Raju (North): 12 customers. Vijay
(South): 10 customers. Total active: 22/100. 3 customers on pause. Have
a great day!
```

Delivery Boy Route Start:

```
Route Today: 12 customers. Start: C001 - Ravi Sharma, 42 Lotus St.
Reply DONE [CODE] or SKIP [CODE] or ISSUE [CODE] [reason]
```

Customer Delivery Confirmation:

```
Your milk was delivered at 6:47 AM today. Enjoy! - Dharma Farms
```

Customer Skip Confirmation:

```
We missed you today. Your subscription extends by +1 day. Next
delivery: tomorrow. - Dharma Farms
```

Customer Pause Confirmation:

```
Paused for 3 days (Jan 5-7). Resume Jan 8. +3 days added to your
subscription. No deliveries during pause. - Dharma Farms
```

Appendix B: Glossary

MVP: Minimum Viable Product - the simplest version that delivers core value

Webhook: Automated message sent from one system to another when an event occurs

API: Application Programming Interface - how software systems talk to each other

UAT: User Acceptance Testing - final testing with real users before launch

PK/FK: Primary Key / Foreign Key - database relationship identifiers

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This Software Requirements Document is a living document. Requirements may be refined as the project progresses through discovery and development phases.